

Exercises: Indices and Data Aggregation

You can check your solutions in the [Judge system](#).

Mr. Bodrog is a greedy small goblin who is in charge of **Gringotts** – the biggest wizard bank. His most precious possession is a small database of the deposits in the wizard's world. Taking money is his hobby. He wants your money as well, but unfortunately, you are not a wizard. The only magic you know is how to work with **databases**. That's how you got access to the precious data. Mr. Bodrog wants you to send him some reports, otherwise he will send a pack of hungry werewolves after you. You don't want to confront a pack of hungry werewolves, do you?

Before going on the next task make sure to download the **Gringotts** database.

Part I – Queries for Gringotts Database

1. Records' Count

Import the database and send the **total count of records** from the one and only table to Mr. Bodrog. Make sure nothing gets lost.

Example

Count
162

2. Longest Magic Wand

Select the **size** of the **longest magic wand**. Rename the new column appropriately.

Example

LongestMagicWand
31

3. Longest Magic Wand Per Deposit Groups

For wizards in **each deposit group** show the **longest magic wand**. Rename the new column appropriately.

Example

DepositGroup	LongestMagicWand
Blue Phoenix	31
...	...

4. Smallest Deposit Group Per Magic Wand Size

Select the **two deposit groups** with the **lowest average wand size**.

Example

DepositGroup
Troll Chest

5. Deposits Sum

Select **all deposit groups** and their **total deposit sums**.

Example

DepositGroup	TotalSum
Blue Phoenix	819598.73
Human Pride	1041291.52
...	...

6. Deposits Sum for Ollivander Family

Select **all deposit groups** and their **total deposit sums**, but **only for the wizards**, who have their magic wands **crafted by the Ollivander family**.

Example

DepositGroup	TotalSum
Blue Phoenix	52968.96
Human Pride	188366.86
...	...

7. Deposits Filter

Select **all deposit groups** and their total deposit sums, but **only for the wizards**, who have their magic wands **crafted by the Ollivander family**. **Filter total deposit amounts lower than 150000**. Order by **total deposit amount** in **descending** order.

Example

DepositGroup	TotalSum
Troll Chest	126585.18
...	...

8. Deposit Charge

Create a query that selects:

- **Deposit group**
- **Magic wand creator**
- Minimum **deposit charge** for each group

Select the data in **ascending** order by **MagicWandCreator** and **DepositGroup**.

Example

DepositGroup	MagicWandCreator	MinDepositCharge
Blue Phoenix	Antioch Peverell	30.00
...	...	...

9. Age Groups

Write down a query that creates 7 different groups based on their **age**.

Age groups should be as follows:

- [0-10]
- [11-20]
- [21-30]
- [31-40]
- [41-50]
- [51-60]
- [61+]

The query should return

- **Age groups**
- **Count** of wizards in it

Example

AgeGroup	WizardCount
[11-20]	21
...	...

10. First Letter

Create a query that returns **all the unique wizard first letters** of their **first names** only if they have **deposit of type Troll Chest**. Order them **alphabetically**. Use **GROUP BY** for uniqueness.

Example

FirstLetter
A
...

11. Average Interest

Mr. Bodrog is highly interested in profitability. He wants to know the **average interest** of all **deposit groups**, split by whether the deposit has **expired or not**. But that's not all. He wants you to select deposits with **start date after 01/01/1985**. Order the data **descending** by **Deposit Group** and **ascending** by **Expiration Flag**.

The output should consist of the following columns:

Example

DepositGroup	IsDepositExpired	AverageInterest
Venomous Tongue	0	16.698947
...	...	

12. *Rich Wizard, Poor Wizard

Mr. Bodrog definitely likes his werewolves more than you. This is your last chance to survive! Give him some data to play his favorite game Rich Wizard, Poor Wizard. The rules are simple:

You **compare** the **deposits of every wizard** with the **wizard after him**. If a wizard is the last one in the database, simply **ignore it**. In the end you have to **sum** the **difference between the deposits**.

Host Wizard	Host Wizard Deposit	Guest Wizard	Guest Wizard Deposit	Difference
Harry	10 000	Tom	12 000	-2000
Tom	12 000	...	...	...

At the end your query should return a **single value**: the SUM of all differences.

Example

SumDifference
44393.97

Part II – Queries for SoftUni Database



That's it! You no longer work for Mr. Bodrog. You have decided to find a proper job as an analyst in SoftUni. It's not a surprise that you will use the **SoftUni** database. Things get more exciting here!

13. Departments Total Salaries

Create a query that shows the **total sum** of **salaries** for **each department**. **Order** them by **DepartmentID**. Your query should return:

- DepartmentID

Example

DepartmentID	TotalSalary
1	241000.00
...	...

14. Employees Minimum Salaries

Select the **minimum salary** from the **employees** for **departments** with ID (2, 5, 7) but **only** for those, **hired after** 01/01/2000.

Your query should return:

- DepartmentID

Example

DepartmentID	MinimumSalary
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2	25000.00
5	12800.00
...	...

15. Employees Average Salaries

Select all **employees** who earn **more than 30000** into a **new table**. Then **delete** all **employees** who have **ManagerID = 42** (in the new table). Then **increase the salaries** of all employees with **DepartmentID = 1** by 5000. Finally, select the **average salaries** in **each department**.

Example

DepartmentID	AverageSalary
1	45166.6666
...	...

16. Employees Maximum Salaries

Find the **max salary** for **each department**. **Filter** those, which have **max salaries NOT** in the **range 30000 – 70000**.

Example

DepartmentID	MaxSalary
2	29800.00
...	...

17. Employees Count Salaries

Count the salaries of all **employees**, who don't have a **manager**.

Example

Count
4

18. *3rd Highest Salary

Find the **third highest salary** in **each department** if there is such.

Example

DepartmentID	ThirdHighestSalary
1	36100.00
...	...

19. **Salary Challenge

Create a query that returns:

- **FirstName**
- **LastName**
- **DepartmentID**

Select all **employees** who have salary **higher than the average salary** of their respective **departments**. Select **only** the first **10 rows**. **Order** them by **DepartmentID**.

Example

FirstName	LastName	DepartmentID
Roberto	Tamburello	1
...	...	